

Blake Werner

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Education

California Institute of Technology

PhD, Mechanical Engineering, Advisor: Dr. Aaron D. Ames
MS, Mechanical Engineering

Pasadena, CA

2025-Present
2023-2025

University of California, Berkeley

BS, Electrical Engineering and Computer Science (EECS) and Mechanical Engineering

Berkeley, CA

2019-2023

Awards and Honors

NSF GRFP Honorable Mention

Spring 2025

Shang-Li and Betty Huang Fellowship

Fall 2023

Caltech Institute Fellowship (MCE)

Fall 2023

UC Berkeley Dean's List

2019-2023

EECS Honors Program

Spring 2023

Edward Frank Kraft Scholarship

Spring 2020

Publications (*Denotes equal contribution)

***Blake Werner**, *Lizhi Yang, Aaron D. Ames. "Architecture Is All You Need: Diversity-Enabled Sweet Spots for Robust Humanoid Locomotion". *IEEE International Conference on Robotics and Automation (ICRA)*. 2026. (In Review)

Pol Mestres, **Blake Werner**, Ryan K. Cosner, Aaron D. Ames. "Probabilistic Control Barrier Functions: Safety in Probability for Discrete-Time Stochastic Systems". *American Control Conference (ACC)*. 2026. (In Review)

*Lizhi Yang, ***Blake Werner**, Massimiliano de Sa, Aaron D. Ames. "CBF-RL: Safety Filtering Reinforcement Learning in Training with Control Barrier Functions". *IEEE International Conference on Robotics and Automation (ICRA)*. 2026. (In Review)

Lizhi Yang, **Blake Werner**, Adrian B. Ghansah, and Aaron D. Ames. "Bracing for Impact: Robust Humanoid Push Recovery and Locomotion with Reduced Order Models". *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*. 2025.

Ryan M Bena, Gilbert Bahati, **Blake Werner**, Ryan K Cosner, Lizhi Yang, Aaron D Ames. "Geometry-Aware Predictive Safety Filters on Humanoids: From Poisson Safety Functions to CBF Constrained MPC". *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*. 2025. (**Best Paper Finalist**)

*Lizhi Yang, ***Blake Werner**, Ryan K. Cosner, David Fridovich-Keil, Preston Culbertson, and Aaron D. Ames. "SHIELD: Safety on Humanoids via CBFs In Expectation on Learned Dynamics". *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2025.

Workshops

Alice M. Agogino Douglas Hutchings, Oliver Parkin, Nomena Randriama, **Blake Werner**. "Multi-Cable Rolling Locomotion with Compliant Parallel Tensegrity Robots". *IEEE International Conference on Robotics and Automation (ICRA)*. 2022.

Research Experience

Advanced Mechanical Bipedal Experimental Robotics Lab

Graduate Student Researcher

Pasadena, CA

2023-Present

- Focus on structured design and safety of control systems for humanoid robots
- Working with the Unitree G1 and Go2 robots, implementing a reinforcement learning pipeline to perform perception-conditioned locomotion over unstructured terrains

- Demonstrated full-stack humanoid control pipeline to sponsors; included autonomous stair climbing, navigation using Fast-LIO, and obstacle avoidance using CBFs

Berkeley Emergent Space Tensegrity Lab

Undergraduate Researcher

Berkeley, CA

2020-2023

- Worked with Mechanical Engineering graduate students to design control software for mobile tensegrity robot using model predictive control; represented team in presenting to ICRA and NASA Ames
- Utilized sparse actuator selection to optimize MPC for various tasks including speed, incline traversal, or power efficiency

Professional Experience

Pebble Mobility Inc.

Robotics Intern

Sunnyvale, CA

Summer 2023

- Designed and wrote safety-critical autonomous features for small electric RV startup
- Prototyped vehicle dynamics model, control algorithms, and path planners in Simulink, then implemented these in C on vehicle GPU where it runs on the current product
- Worked with small team to integrate autonomous stack into NVIDIA DriveWorks compute graph framework (CGF)

Berkeley Robot Open Autonomous Racing Team

Estimation Project Lead

Berkeley, CA

Spring 2023

- Led team of four students to design Extended Kalman Filter for state estimation for autonomous F1 race car
- Utilized Dugoff Tire Model combined with 11-state car dynamics and data from high-fidelity simulation to derive and tune model and estimator
- Validated estimator both in sim and on Dallara AV-21 vehicle, where it was used in Milano Monza Motor Show 2023, and all races since

Combat Robotics at Berkeley

EECS Drive Member

Berkeley, CA

2021-2023

- Designed, sourced parts for, and tested electronic speed controllers (ESCs), batteries, motors, and radio transmitters and receivers for the drive system for Glitch, as seen on Battlebots show (**Rookie of the Year, Battlebots Season VI**)
- Tuned ESCs for performance and power draw subject to motor torque curve and battery constraints
- Worked with the mechanical drive team to design hardware using Fusion360; personally manufactured parts with water jet, CNC mill, and lathe; and integrated components into the final product

Hello Robot Inc.

Robotics Intern

Martinez, CA

Summer 2022

- Designed simple optimization-based model and controller for stepper motor to improve tracking under transient disturbances
- Used Python and Arduino C to control and directly interface with the robot, and Matlab Simulink to prototype and gather data

Politiq Inc.

Cofounder/CTO of Political Software Startup

Berkeley, CA

2019-2020

- Hired employees, led design and implementation of minimum viable product, a working website with SQL backend
- Presented and pitched finished work, leading Politiq to win a spot at the Berkeley Skydeck Startup Incubator

Teaching Experience

Caltech Mechanical and Civil Engineering Department

Teaching Assistant for ME169 - Mobile Robots

Pasadena, CA

Spring 2025

- Students start with a blank Raspberry Pi and custom robot hardware, and end with a fully functioning mobile robot performing custom-built SLAM in a timed game and an unknown environment
- Taught concepts regarding, and assisted in implementation of, low level encoder design and motor feedback, Lidar odometry, and kinematic and dynamic planning

Caltech Computing and Mathematical Sciences Department

Teaching Assistant for CDS110 - Analysis and Design of Feedback Control Systems

Pasadena, CA

Fall 2024

- Taught concepts such as linear systems theory, stability analysis, optimal control, and reinforcement learning
- Created assignments such as an optimal control/MPC project in which students learn CasADi and use it to program an optimal controller for a simple bicycle model to perform obstacle avoidance and planning

UC Berkeley Mechanical Engineering Department

Head Graduate Student Instructor for ME100 - Electronics for IOT

Berkeley, CA

2022-2023

- Taught concepts such as circuit design, data processing, and filtering; with applications such as sensors, actuators, microcontrollers, and firmware

Project Highlights

Sawyer Dart Thrower

Berkeley, CA

Robotics

Fall 2022

- Designed and implemented hardware and software allowing Sawyer robot to track targets and hit them with darts
- Used computer vision (canny edge detection) to find target, and determine orientation and distance
- Developed software for Sawyer to position itself using inverse kinematics, and a custom launcher allowing it to shoot the dart, reliably hitting the target

Quadruped Mechanism

Berkeley, CA

Planar Machinery

Fall 2021

- Designed and built a quadruped walking robot with variable gaits and turning capabilities
- Used Grashof linkage design to allow for minimal degrees of freedom in the legs
- Visualized design using CAD (Solidworks), then constructed using machined Al6061 parts

Skills and Interests

Languages: Python, C++, C/CUDA, MATLAB

Robotics Stack: IsaacLab, Mujoco, ROS, ROS2, Unitree SDK

Perception/Estimation: VIO/SLAM (Fast-LIO, LIO-SAM), Sensor calibration, fusion, and interface (MID360 Lidar, ZED2, Intel Realsense D435i, Intel Realsense t265)

Design: SolidWorks, Fusion360, Machine shop (CNC, TIG, Mill, Lathe)